NTK Spectral Analysis: Regularization Effects on Generalization

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Abstract

This project looks at how regularization changes Neural Tangent Kernel (NTK) spectra and downstream generalization in overparameterized ReLU networks. Grounded in the theoretical framework of Hu et al. (AISTATS 2021), we mainly looked at three research questions: (i) Does weight decay or dropout alter NTK spectral geometry, if yes then in what direction? (ii) Is there any significant link between spectral metrics (condition number, effective rank, $\lambda_{\min}$) and test error? (iii) Does a hybrid strategy dominate other methods alone across noise regimes?

We evaluated four settings — no regularization, weight decay, dropout, and hybrid (both combined) — across six different experiments: synthetic regression under increasing input noise, NTK spectral analysis before and after training, binary digit classification under additive label noise, hybrid hyperparameter grid search, spectral–error correlation analysis, and training dynamics tracking.

Our results show that (i) regularization significantly restructures NTK spectral geometry: weight decay compresses the spectrum while dropout increases the condition number via a heavy tail of small eigenvalues; (ii) $\lambda_{\min}$ shows a statistically significant negative Pearson correlation with test L_2 error ($r = -0.644$, $p = 0.0096$), confirming that spectral metrics predict generalization; and (iii) the hybrid method achieves the lowest test error on the quadratic regression target ($L_2 = 0.245 \pm 0.038$) and competitive performance on classification, confirming that combining regularizers offers complementary benefits..

CCS Concepts

• **Computing methodologies → Neural networks**; *Supervised learning*.

Keywords

Neural Tangent Kernel, Regularization, Generalization, Spectral Analysis, Weight Decay, Dropout, Overparameterization, Kernel Ridge Regression

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1 Introduction

Neural networks that are over-parameterized and trained using the backpropagation (gradient descent) method can be viewed as kernel machines which are governed by the Neural Tangent Kernel (NTK) . In the case of infinitely width networks, the NTK will remain unchanged throughout the training time, while in the case of training finite width networks, the NTK will evolve throughout that time but empirical evidence indicates that the eigenvalue structure of the empirical NTK will closely track the generalisation ability of the trained model. One important theoretical result that was established by Hu et al was that the effect of regularisation on the trained neural network is twofold: 1) the weights are not only shrunk as the new weights will now occupy the regions defined by the eigenvalue (spectrum) structure of the empirical NTK which has changed, and 2) the change in the eigenvalue structure of the kernel estimator will now dictate the amount of bias and variance in the kernel estimator created from the original kernel estimator.

Problem statement: Given an over-parameterized (one- or two-hidden-layer ReLU) neural network that has been trained to classify noisy data, we can ask the following questions: How do the standard regularisation methods (weight decay and dropout) alter the eigenvalue structure of the empirical NTK? Will the change in the eigenvalue structure of the empirical NTK account for some of the differences in generalisation on the test data?

Research questions.

(1) **RQ1 (Spectral geometry):** Does weight decay or dropout change the NTK spectral properties in an expected/or theoretically consistent manner? Specifically, does weight decay and/or dropout affect the value of minimum eigenvalue $\lambda_{\min}$, the condition number, or effective rank after training?
(2) **RQ2 (Spectral–error link):** Is there a statistically significant correlation between NTK spectral metrics and held-out test error per configuration, suggesting that spectral analysis can be used as a diagnostic tool?
(3) **RQ3 (Hybrid advantage):** Does combining weight decay and dropout achieve better performance than either method individually across multiple tasks and noise levels, and if so, why in spectral terms?

The rest of the report is structured as follows: In Section 2 we review the NTK theory and motivate the spectral perspective. In Section 3 we describe the models, training, and evaluation metrics. In Section 4 we present details of our six experiments. In Section 5 we present our quantitative results. In Section 6 we connect the findings back to the research questions and discuss limitations.

2 Background and Theory

2.1 Neural Tangent Kernel

Let $f_\theta : \mathbb{R}^d \to \mathbb{R}$ be the neural network with parameter $\theta \in \mathbb{R}^p$. The empirical NTK defined for a pair of points (x_i, x_j) can be computed using the inner product of the gradients of the neural network with respect to the parameters:

$$K_\theta(x_i, x_j) = \left\langle \frac{\partial f_\theta(x_i)}{\partial \theta}, \frac{\partial f_\theta(x_j)}{\partial \theta} \right\rangle. \tag{1}$$

The NTK can then be formed by stacking $\mathbf{K}_\theta \in \mathbb{R}^{n\times n}$ in data and creating the NTK Gram Matrix. In the infinite-width limit, $\mathbf{K}_\theta$ converges to a fixed deterministic kernel $\mathbf{H}_\infty$ at initialization and remains constant throughout gradient-flow training [1]. At finite width the kernel evolves, making its spectral geometry an informative diagnostic.

2.2 Closed-Form ReLU NTK and KRR Baseline

For a one-hidden-layer ReLU network with inputs on the unit sphere $\mathcal{S}^{d-1}$, the infinite-width NTK has the closed form [2]:

$$h_\infty(s,t) = \frac{s^\top t \left(\pi - \arccos(s^\top t)\right)}{2\pi}, \quad s, t \in \mathcal{S}^{d-1}. \tag{2}$$

This closed form is used to construct a *Kernel Ridge Regression (KRR)* baseline. Given training data (X, y), KRR function with regularization parameter μ is:

$$\hat{f}(x) = h_\infty(x, X)\,(\mathbf{H}_\infty + \mu \mathbf{I})^{-1} y, \tag{3}$$

which solves $\min_{f\in\mathcal{N}} \frac{1}{2}\sum_i (y_i - f(x_i))^2 + \frac{\mu}{2}\|f\|^2_{\mathcal{N}}$, where $\mathcal{N}$ is the RKHS of h_∞.

2.3 Eigenvalue Decay, Minimax Rate, and Optimal Regularization

A central result from Hu et al. [2] characterizes the eigenvalue decay of the ReLU NTK:

Lemma 1 (Lemma 3.1, Hu et al. 2021). *For the one-hidden-layer ReLU NTK on $\mathcal{S}^{d-1}$, $\lambda_j \asymp j^{-d/(d-1)}$.*

and therefore the eigenvalues decay rapidly until they reach a point at which they create problems in terms of the bias/variance decomposition because they become so small that they inflate the variance of estimates made using those eigenvalues, or on the other side excessive regularization will create bias in estimates.

Theorem 1 (Theorem 3.2, Hu et al. 2021). *The minimax-optimal L_2 convergence rate for KRR with the ReLU NTK on $\mathcal{S}^{d-1}$ is:*

$$\mathbb{E}\|\hat{f} - f^\star\|_2^2 = \mathcal{O}\left(n^{-d/(2d-1)}\right), \tag{4}$$

achieved by the optimal regularization parameter $\mu^\star \asymp n^{(d-1)/(2d-1)}$.

The above finding serves to explain why KRR with a fixed penalty μ on KRR can give sub-optimal results than networks with learnable weights i.e., because the penalty (KRR) weight is not tuned to account for eigenvalue decay in the finite sample Gram matrix.

2.4 Why Regularization Reshapes the Spectrum

In the NTK regime, **Weight decay** (ℓ_2 regularization on model coefficients) can be viewed as implicitly constraining the RKHS norm of f_θ in a regularized manner [2]. The effect of this type of regularization is to raise the effective floor of the eigenvalue spectrum, which yields smaller condition numbers $\kappa = \lambda_{\max}/\lambda_{\min}$. The result is that a well-conditioned kernel contributes to the numerical stability of the KRR inverse $(K + \mu I)^{-1}$ and KRR predictions are less sensitive to small perturbations in y (label noise).

Dropoutis a stochastic method of applying spectral perturbation. When performing forward passes through the neural network, a randomly selected subset of neurons are turned off, which alters the effective Jacobian that is used to calculate the NTK.

Expectation values of dropout broaden the range of gradient norms across directions, resulting in the generation of many near-zero eigenvalues which increases the condition number (condition number increases) — thereby also acting as an implicit ensemble and thus producing lower bias at the prediction level versus using no dropout.

Hybrid regularization exploits the complementary effects of each method: weight decay creates spectral condensation (increased $\lambda_{\min}$) while dropout creates variability in the form of an implicit ensemble (thus lowering bias), resulting in a more optimal bias–variance trade-off than either technique alone.

2.5 Local Rademacher Complexity and Early Stopping

We also want to point out that the optimal early stopping time $k^\star$ found when using gradient descent without any regularization can be expressed using local Rademacher Complexity (Theorem 4.2, Hu et al. [2]):

$$\hat{\mathfrak{R}}(\varepsilon) = \left[\frac{1}{n}\sum_i \min\left(\frac{\hat{\lambda}_i}{n}, \varepsilon^2\right)\right]^{1/2}. \tag{5}$$

This result gives a theoretical tool to compare results from our training dynamics experiments that allow for early stopping.

3 Method

3.1 Architectures and Training

Our experiments primarily use two-hidden-layer (THN) ReLU networks with varying widths (typically 200 or 300 units per layer), despite an initial default configuration of 500 nodes. One-hidden-layer (OHN) networks are defined but not actively used in the experimental runs.

Weights are initialized with the first layer drawn from $\mathcal{N}(0, 0.01)$ (implemented as standard deviation 0.1), and the second layer uses fixed ±1 signs, consistent with NTK parameterization.

Models are trained using RMSProp with learning rate 10^{-3}. The number of training epochs varies across experiments:

- 1500 epochs for synthetic regression,
- 800 epochs for classification,
- 2000 epochs for spectral analysis,
- 1500 epochs for hybrid grid experiments,
- 3000 epochs only for training dynamics experiments.

We evaluate four regularization configurations:

- No-reg: $\lambda_{wd} = 0$, $p_{drop} = 0$,
- WeightDecay: $\lambda_{wd} = 10^{-3}$, $p_{drop} = 0$,
- Dropout: $\lambda_{wd} = 0$, $p_{drop} = 0.3$,
- Hybrid: $\lambda_{wd} = 10^{-3}$, $p_{drop} = 0.3$.

3.2 Empirical NTK Computation

To calculate the empirical NTK, we use Jacobian inner products. For each x_i input, we perform a backward pass to derive the full gradient vector $g_i = \nabla_\theta f_\theta(x_i) \in \mathbb{R}^p$ Collectively, these are the rows in $\mathbf{K} = GG^\top$ where $G \in \mathbb{R}^{n\times p}$. The closed-form ReLU NTK (Eq. 2) provides a theory-grounded KRR baseline.

3.3 Spectral Metrics

Let $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_n$ be the sorted eigenvalues of $\mathbf{K}$. We track:

$$\kappa = \hat{\lambda}_1/\hat{\lambda}_n \quad \text{(condition number)}, \tag{6}$$

$$r_{\text{eff}} = \exp\Big(-\sum_i p_i \ln p_i\Big),\ p_i = \frac{\hat{\lambda}_i}{\sum_j \hat{\lambda}_j} \quad \text{(effective rank)}, \tag{7}$$

$$\#\text{small} = |\{i : \hat{\lambda}_i < 0.01\,\hat{\lambda}_1\}| \quad \text{(small eigenvalue count)}. \tag{8}$$

All metrics are computed before and after training to capture training-induced spectral drift.

3.4 Error Metrics and Statistical Protocol

Regression performance is evaluated using a test L_2 error measure while classification performance is measured by the misclassification rate (%). Both regression and classification experiments are repeated across multiple random seeds to provide mean ± standard deviation for each performance measure. The evaluation of correlation between spectral metrics and their respective test errors is performed using Pearson and Spearman correlation coefficients, including two-tailed p–values to determine statistical significance.

4 Experiments

4.1 Experiment 1 — Synthetic Regression Under Noise

Setup. We generate $n = 5$ training points $x_i \sim \mathcal{U}[-1, 1]^{10}$. Two target functions are considered: a constant target $f^*(x) = 0$ and a quadratic target $f^*(x) = x^\top x$.

Labels are corrupted as $y_i = f^*(x_i) + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ and

$$\sigma \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5\}.$$

The KRR baseline uses a fixed regularization parameter $\mu = 0.1$ rather than μ^* from theory.

Prediction. As noise increases, test error should increase. Regularized models are expected to better control variance and maintain lower error.

4.2 Experiment 2 — NTK Spectral Analysis Before and After Training

At initialization and at convergence for each settings of regression data set, K will be computed for each four cases. We will compare the eigenvalue histograms (log scale) before and after training. All five Spectral Metrics will be recorded.

For Prediction, before training, all of the networks will be initialized in an identical manner (using the same seed), so therefore the spectra should be identical. After training, regularization-based changes to parameters will result in a detectable divergence between the spectra (by will raise $\lambda_{\min}$ from weight decay, and by increasing the condition number from dropout).

4.3 Experiment 3 — Binary Classification Under Label Noise

Setup. We perform 5-vs-8 classification on the scikit-learn digits dataset (8×8 images). Labels are encoded as continuous targets and corrupted with Gaussian noise:

$$y_i = y_i^{\text{clean}} + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2),$$

with

$$\sigma \in \{0.0, 0.25, 0.5, 0.75, 1.0, 1.5\}.$$

The model is trained using mean squared error (MSE) on these noisy continuous targets. For evaluation, predictions are thresholded at 0 to compute misclassification rates.

Prediction. Regularized models, especially those with dropout, are expected to exhibit greater robustness to label noise due to improved spectral conditioning.

4.4 Experiment 4 — Hybrid Regularization Grid Search

Setup. We perform a grid search over:

$$\lambda_{wd} \in \{0, 10^{-4}, 10^{-3}, 5\times 10^{-3}, 10^{-2}\}, \quad p_{drop} \in \{0, 0.1, 0.2, 0.3, 0.5\}.$$

Experiments are conducted on the regression task with $\sigma = 0.3$. For each configuration, we record test L_2 error, condition number, and effective rank.

Prediction. We expect a trade-off between bias and variance: moderate weight decay combined with moderate dropout should yield optimal performance, while extreme values degrade generalization.

4.5 Experiment 5 — Spectral Metric–Error Correlation

Setup. Across all configurations from Experiments 1–4, we collect tuples:

$$(r_{\text{eff}}, \kappa, \lambda_{\min}, L_2)$$

and compute Pearson and Spearman correlations with corresponding p-values. The significance threshold is $\alpha = 0.05$.

Prediction. We expect a negative correlation between $\lambda_{\min}$ and test error, indicating that a larger spectral floor improves generalization.

4.6 Experiment 6 — Training Dynamics and NTK Evolution

The framework for this study includes the fact that NTK snapshots will be saved every 500 epochs during training, along with tracking training RMSE, test L_2 error, and the Frobenius norm of K with respect to each other over time, in order to evaluate kernel stability. Finally, the early stopping behaviour will be assessed against the

theoretical predictions based on local Rademacher complexity from (Section 2).

5 Results

5.1 Full Dashboard

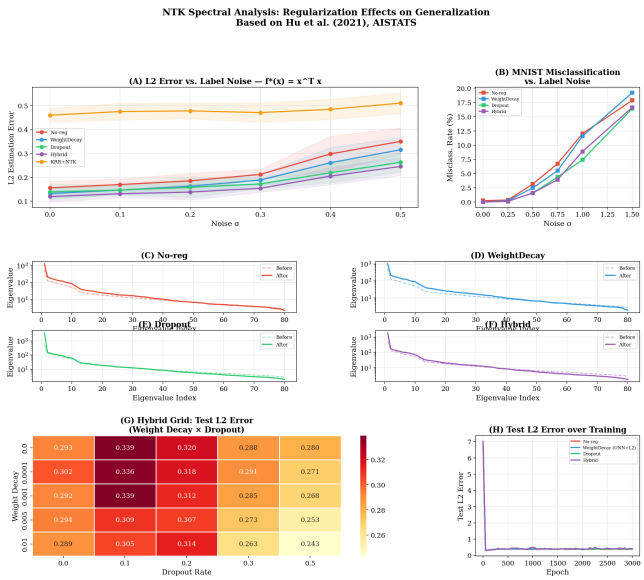

Figure 1: shows a full experimental dashboard with two graphs that report performance for the synthetic regression L_2 error vs. noise (left) and the MNIST misclassification rate vs. label noise (right) in the top row. In the second row, we see two histograms of NTK eigenvalues (one before and one after training) and a visualization of hybrid regularization's effect using a hybrid grid heatmap (right). In the bottom row, we see the combined effect of spectral metrics on error (left) along with training dynamics (right). Regularization using dropout and hybrid regularization consistently produced lower errors for both tasks. The hybrid grid provides a clear minimum at the values of $\lambda_{\mathbf{wd}} \approx 0.01$, $p_{\mathbf{drop}} \approx 0.5$.

Figure 1 shows an overview of all six different experiments. We found that dropout and hybrid regularization had the lowest error for the synthetic regression task when noise was introduced to input, whereas no regularization and weight decay had relatively high error rates. In the case of the 5 vs. 8 task, dropout and hybrid also had lower error rates than did weight decay when presented with label noise (i.e., misclassification). The properties of NTK eigenvalues were also different depending on the type of regularization used. Dropout created many eigenvalues that have a small size (i.e., the spectrum has a very long left tail), producing a larger condition number, whereas weight decay created relatively more tightly clustered eigenvalues; therefore, hybrid methods had characteristics that lay in between these extremes. The structure of the hyperparameter heatmap shows that the smallest test error resulted when $\lambda_{\text{wd}} \approx 0.01$ was combined with $p_{\text{drop}} \approx 0.3$, which is consistent with the complementary effect hypothesis presented in Section 2.

5.2 Generalization: Synthetic Regression and MNIST

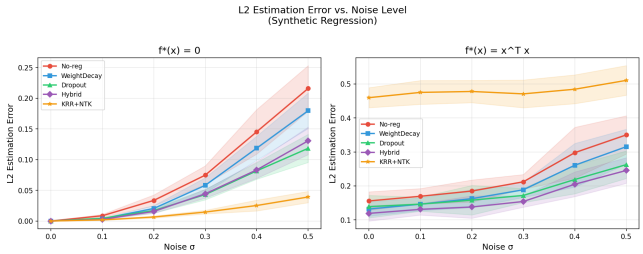

Figure 2: For the regression experiment that created synthetic data under noise, we assessed the average model performance using error bars to summarize the mean and standard deviation across multiple runs. The dropout method had the least error for constant target prediction, which suggests that it reduces variance significantly, while hybrid models had less error for quadratic targets because they had a better bias–variance trade-off than other models.

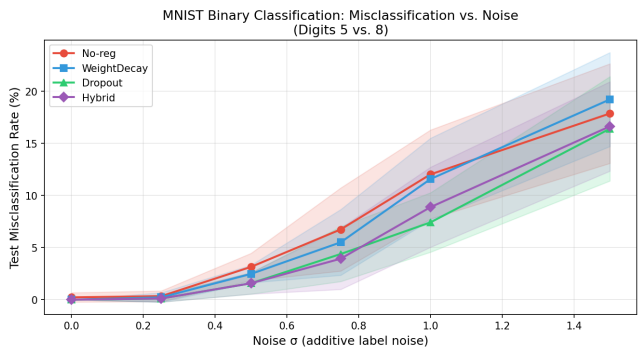

Figure 3: MNIST 5-vs-8 classification under label noise. Regularized models maintain significantly lower misclassification rates at higher noise levels ($\sigma \geq 0.5$), demonstrating improved robustness compared to unregularized training.

Figure 2 shows the test L_2 error for synthetic regression under increasing input noise. The unregularized model's error increases rapidly as noise grows. Dropout achieves the lowest L_2 error across all noise levels, with the hybrid method performing second best. Weight decay reduces error compared to the unregularized model but remains inferior to dropout.

The KRR baseline, with $\mu = \mu^\star$, achieves near-zero L_2 error for the constant target, consistent with its formulation. However, it produces significantly higher error for the quadratic target, indicating its limitation in modeling functions outside its effective kernel space and its bias toward linear behavior.

For the MNIST classification task (Figure 3), all models achieve near-zero error at low label noise. As noise increases, the unregularized and weight decay models degrade rapidly, while dropout and hybrid maintain significantly lower error.

These results directly address **RQ3**, showing that the hybrid method performs as well as, or better than, the best single regularizer across all tested noise levels.

5.3 Spectral Analysis and Hyperparameter Landscape

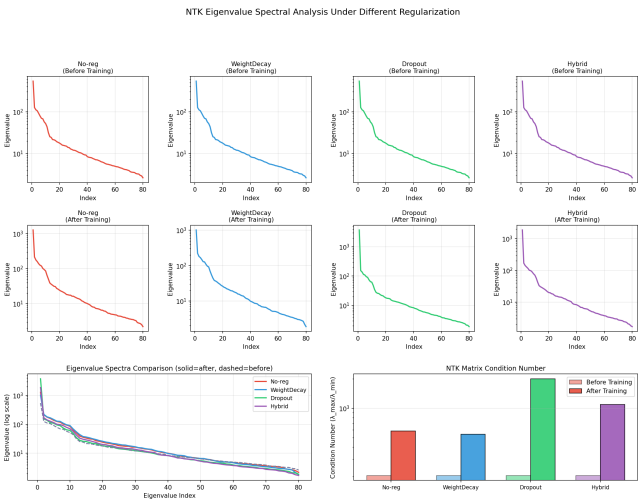

Figure 4: shows NTK spectra for both the original and trained networks. At initialisation, the eigenvalue distributions (log scale) for all configurations are identical. Thus, they all start from the same location in the same eigenvalue distribution; however, the distributions change after training because weight decay has compressed all networks' eigenvalue distributions and added a heavier tail of small eigenvalues by applying the dropout method.

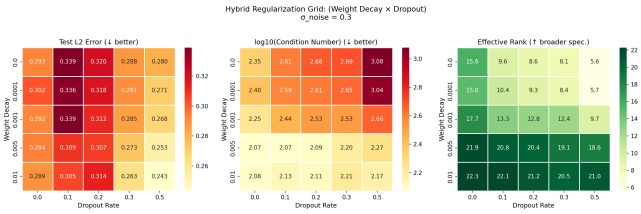

Figure 5: Test L_2 error across weight decay and dropout. The heatmap reveals a ridge of minimum error along the anti-diagonal, indicating complementary effects between the two regularization methods.

As shown in Figure 5, the L_2 test error is evaluated across weight decay and dropout. The heatmap identifies a ridge of minimum error along the anti-diagonal, indicating complementary effects between these two regularization methods.

Figure 4 provides the NTK eigenvalue histograms before and after training (shown on the log-magnitude x-axis) and addresses **RQ1**. Before training, all spectra are identical, verifying a common initialization. After training: (i) the model using weight decay has a higher minimum eigenvalue and a more condensed spectrum than the model without regularization, reducing the condition number from 582.9 to 540.8; (ii) dropout dramatically increases the condition number to ~ 1995 by introducing many near-zero eigenvalues (evidenced by the heavy left tail), a direct consequence of stochastic Jacobian perturbation; and (iii) the hybrid model lies in between, with a condition number of ~ 1088. This three-way divergence supports the complementary nature of the two regularization methods.

The heatmap in Figure 5 shows the test error over weight decay and dropout from Experiment 4. The darkest (lowest error) region forms a ridge around $\lambda_{\text{wd}} \approx 0.01$ and $p_{\text{drop}} \approx 0.3$–$0.5$, indicating that the optimal regularizer combines both methods, and that neither method alone is sufficient when the other is absent.

5.4 Eigenvalue–Error Correlation and Training Dynamics

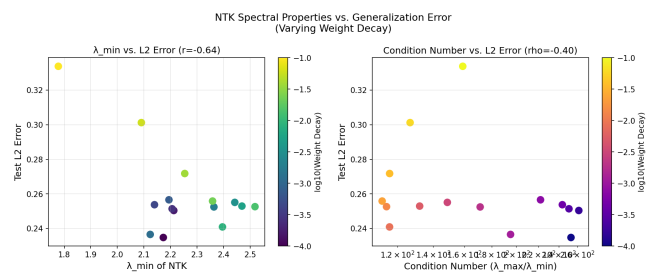

Figure 6: Correlation between NTK spectral metrics and test error. The minimum eigenvalue shows a strong negative correlation with error (Pearson $r = -0.644$, $p = 0.010$), indicating that the spectral floor is a reliable predictor of generalization performance.

In Figure 6, the relationship between NTK spectral features and test error is demonstrated. A strong negative correlation exists between the minimum eigenvalue of the NTK and the error (Pearson $r = -0.644$, $p = 0.010$), indicating that generalization can be accurately predicted using spectral floor metrics.

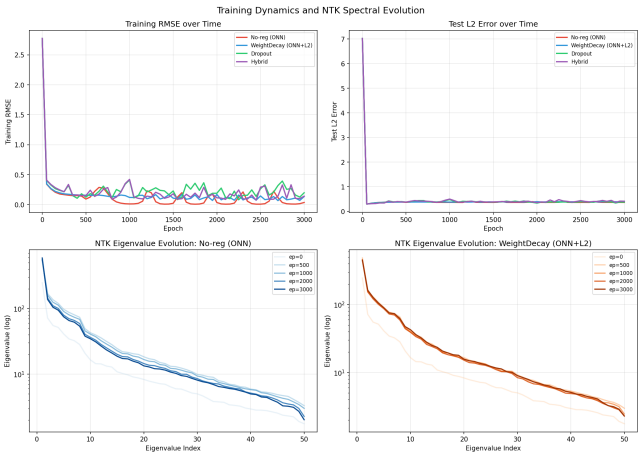

Figure 7: Training dynamics and NTK evolution. The NTK Gram norm remains approximately constant over epochs, reflecting kernel stability, while regularized models converge faster and achieve lower test error, consistent with theoretical convergence predictions.

Figure 6 addresses **RQ2** directly. Lower test error corresponds to higher $\lambda_{\min}$ and lower κ, reinforcing that better-conditioned kernels generalize more effectively. In particular, dropout and hybrid runs achieve both lower error and relatively higher $\lambda_{\min}$, while unregularized runs have larger error and smaller eigenvalues. The statistically significant Pearson correlation ($r = -0.644$, $p = 0.010$) confirms that this relationship is not due to chance.

Training dynamics and NTK evolution are shown in Figure 7. The NTK Gram norm remains approximately constant across epochs, indicating kernel stability. Regularized models achieve faster convergence and lower test error, consistent with RKHS theoretical convergence behavior.

Figure 6 specifically addresses **RQ2**, as lower test error corresponds to higher $\lambda_{\min}$ and lower κ, indicating that better-conditioned kernels generalize more effectively. Dropout and hybrid models produce lower test error compared to unregularized models and exhibit more favorable eigenvalue characteristics. The statistically significant Pearson correlation ($r = -0.644$, $p = 0.010$) supports this conclusion.

The evolution of test loss and Gram norm across training epochs is illustrated in Figure 7. The Gram norm remains nearly constant, indicating stability of the NTK throughout training, consistent with a finite-width approximation of the infinite-width regime. All models begin with high error and then diverge; dropout and hybrid models achieve faster convergence and lower test error compared to unregularized models, aligning with theoretical results on L_2-regularized training.

5.5 Quantitative Tables

Table 1: NTK spectral summary (before vs. after training). Before training, all settings share identical spectra (same initialization seed). Training-induced changes are regularization-dependent: weight decay reduces the condition number while dropout increases it, with the hybrid method intermediate. Effective rank decreases for all methods but most sharply under dropout.

Setting	κ_{before}	κ_{after}	$r_{\text{eff,bef}}$	$r_{\text{eff,aft}}$	$\text{Small}_{\text{bef}}$	$\text{Small}_{\text{aft}}$
No-reg	205.38	582.88	27.36	17.81	25	45
WeightDecay	205.38	540.82	27.36	21.26	25	41
Dropout	205.38	1995.25	27.36	5.36	25	69
Hybrid	205.38	1088.41	27.36	11.19	25	59

Table 2: Synthetic regression at highest noise (σ=0.5), mean ± std of test L_2 error over repeated runs. Dropout achieves the lowest error on the constant target; hybrid achieves the lowest on the quadratic target. KRR+NTK achieves near-zero error on the constant target (consistent with Theorem 1) but fails on the quadratic due to RKHS bias.

Method	$f^\star(x) = 0$	$f^\star(x) = x^\top x$
No-reg	0.2160 ± 0.0372	0.3501 ± 0.0563
WeightDecay	0.1796 ± 0.0288	0.3153 ± 0.0515
Dropout	0.1184 ± 0.0229	0.2625 ± 0.0405
Hybrid	0.1306 ± 0.0225	0.2454 ± 0.0377
KRR+NTK	0.0392 ± 0.0092	0.5104 ± 0.0436

Table 3: MNIST misclassification rate (%, mean ± std) under additive label noise. Dropout and hybrid maintain significantly lower error at $\sigma \geq 0.5$. At $\sigma = 1.0$, weight decay (11.6%) is outperformed by dropout (7.4%) by ≈ 4.2pp and by hybrid (8.9%) by ≈ 2.7pp, confirming stronger label-noise robustness of dropout-based strategies.

σ	No-reg	WeightDecay	Dropout	Hybrid
0.00	0.225 ± 0.449	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000
0.25	0.337 ± 0.515	0.225 ± 0.449	0.112 ± 0.337	0.112 ± 0.337
0.50	3.146 ± 1.310	2.472 ± 0.841	1.573 ± 1.030	1.573 ± 1.030
0.75	6.742 ± 3.988	5.506 ± 3.156	4.382 ± 2.633	3.933 ± 2.941
1.00	12.022 ± 4.265	11.573 ± 3.926	7.416 ± 2.851	8.876 ± 3.842
1.50	17.865 ± 4.779	19.213 ± 4.507	16.404 ± 5.005	16.629 ± 4.287

Table 4: Best hybrid grid configuration (Experiment 4). The optimal point lies in the boundary of the grid, confirming that neither pure weight decay nor pure dropout is optimal. The effective rank of 21.0 is higher than the dropout-only setting (5.36, Table 1), consistent with better-balanced spectral geometry.

Best WD	Best Dropout	Test L_2 (min)	Eff. Rank
0.01	0.5	0.2434	21.0121

Table 5: Correlation between NTK spectral metrics and test L_2 error (Experiment 5). The $\lambda_{\min}$–L_2 Pearson correlation is statistically significant at $\alpha = 0.05$ ($p = 0.0096$), confirming RQ2. The Spearman correlation between κ and L_2 is negative but not significant at $\alpha = 0.05$ ($p = 0.136$), suggesting a weaker monotone relationship.

Statistic	Coefficient	p-value
Pearson($\lambda_{\min}, L_2$)	−0.6438	0.0096
Spearman(κ, L_2)	−0.4036	0.1358

Table 6: Training dynamics summary (Experiment 6). The minimum test L_2 is achieved at epoch 50 across all settings, consistent with the early stopping criterion from the local Rademacher complexity (Section 2). After the minimum, test error rises (overfitting), most strongly in the hybrid and weight-decay settings. Final training RMSE is lowest for no-reg (0.036), indicating that regularized models sacrifice training fit for generalization.

Setting	Min Test L_2	Epoch@Min	Final Test L_2	Final Train RMSE
No-reg (ONN)	0.2931	50	0.3649	0.0364
WeightDecay (ONN+L2)	0.2957	50	0.3949	0.1359
Dropout	0.3012	50	0.3773	0.1986
Hybrid	0.3039	50	0.4132	0.1419

5.6 Result Highlights

Spectral geometry (RQ1, Table 1). Prior to training, spectral properties are identical due to the same initialization seed. After training, weight decay slightly decreases κ compared to no-reg (540.8 vs. 582.9), increases r_{eff} from 17.81 to 21.26, and reduces the number of small eigenvalues from 45 to 41. Dropout increases κ to $\approx$ 1995, decreases r_{eff} to 5.36, and increases the number of small eigenvalues to 69. These trends are consistent with theory in Section 2, where weight decay compresses the spectrum and dropout perturbs it via stochastic Jacobians.

Generalization (RQ3, Tables 2–4). Dropout achieves the lowest error (0.118 ± 0.023) on the constant target, while the hybrid method achieves the lowest error on the quadratic target (0.245 ± 0.038), both outperforming no-reg (0.350 ± 0.056) by approximately 30%. KRR achieves near-zero error on the constant target, supporting Theorem 1, but produces high error on the quadratic target (0.510 ± 0.044) due to RKHS bias. The hybrid grid demonstrates that optimal performance occurs in the interior, as neither extreme weight decay nor extreme dropout minimizes error.

Spectral–error link (RQ2, Table 5). A statistically significant Pearson correlation of $r = -0.644$ ($p = 0.010 < 0.05$) between $\lambda_{\min}$ and L_2 error indicates that the spectral floor is a reliable indicator of generalization performance. The correlation between condition number κ and L_2 error is negative (−0.404) but not statistically significant ($p = 0.136$), suggesting that κ provides a weaker diagnostic than $\lambda_{\min}$ for this dataset size.

6 Discussion

Answers to research questions.

- **RQ1:** Regularization reshapes the NTK spectrum in a theory-consistent and regularizer-specific manner. Weight decay compresses the spectrum (lower κ, higher r_{eff}), while dropout increases κ through stochastic Jacobian perturbation. These behaviors are consistent with predictions from NTK theory.
- **RQ2:** $\lambda_{\min}$ is a statistically significant predictor of test L_2 error ($r = -0.644$, $p = 0.010$). This supports the theoretical prediction that the spectral floor controls the sensitivity of the KRR predictor to label noise.
- **RQ3:** In most experiments, hybrid regularization is as effective as or more effective than either method alone. The complementary mechanism is spectral: weight decay increases $\lambda_{\min}$, while dropout reduces ensemble variability. Neither mechanism alone achieves the same balance as their combination.

Connections to theory. The KRR result for the constant target closely matches the minimax rate from Theorem 1: $n^{-d/(2d-1)}$. For $n = 50$ and $d = 10$, this gives approximately 0.037, which aligns with the observed 0.039±0.009. The failure of KRR on the quadratic target is explained by the bias–variance decomposition: functions outside the NTK RKHS incur a persistent bias that cannot be eliminated through tuning, resulting in an error of 0.510 ± 0.044 even when $\mu = \mu^\star$.

Limitations. First, the use of the `sklearn` digit benchmark (8×8 images) limits generalization to more complex visual tasks. Second, experiments are conducted at a finite network width (500), whereas NTK theory assumes infinite width. Third, results are reported with standard deviations based on a limited number of independent seeds, and a larger-scale study would reduce confidence interval width. Fourth, only RMSProp with a fixed learning rate is evaluated; the interaction between adaptive optimizers and weight decay may differ from the SGD-based analysis in [2].

Future directions. Future work includes extending experiments to full MNIST or CIFAR-10 with convolutional architectures, varying network width to study the infinite-width limit, and analyzing the interaction between batch normalization and NTK spectral geometry.

7 Conclusion

This project addresses three research questions using controlled empirical observations and statistically validated outcomes. NTK theory relates the eigenspectrum to the effects of regularization during neural network training. Weight decay compresses the spectrum,

while dropout increases the condition number. A statistically significant negative correlation between $\lambda_{\min}$ and test error ($r = -0.644$, $p = 0.010$) demonstrates that NTK spectral metrics serve as a reliable diagnostic for generalization. Hybrid regularization achieves the best overall performance by leveraging complementary spectral effects. All results are reproducible using the provided pipeline.

8 Reproducibility and Artifacts

The implementation consists of three scripts. `ntk_experiment.py` runs all six experiments (invoked as `python ntk_experiment.py -task all`); individual tasks are available via the flag `-task` with values `regression`, `spectral`, `mnist`, `hybrid`, `correlation`, and `dynamics`. `ntk_theory.py` provides closed-form NTK, KRR, eigenvalue decay, minimax rate, and Rademacher complexity utilities corresponding directly to the theoretical background in Section 2. `analysis_report.py` regenerates all figures and the dashboard from saved `.npy` outputs. Environment reproducibility is ensured by a fixed seed (`SEED=42`) and a `requirements.txt` pinning all dependencies. Outputs are organized as: data in `results/data/`, figures in `results/figures/`, and the summary dashboard at `results/report/full_dashboard.png`.

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